

Supporting Information

Design of a Flexible, Zn-Selective Protein Scaffold that Displays Anti-Irving–Williams Behavior

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Content	Pages
Methods	S2-S8
Supporting Tables 1-6	S9-S16
Supporting Figures 1-8	S16-S23
References	S26

Methods

Protein Mutagenesis, Expression, and Purification. A pET-20b(+)/AB vector with ampicillin resistance gene was used to prepare the B (^{H67I/H71Q/H97T}AB) construct.¹ Site-directed mutagenesis by polymerase chain reaction (PCR) was performed using PfuUltra high-fidelity DNA polymerase (Agilent). The resulting PCR products were transformed to chemically competent *Escherichia coli* (*E. Coli*) XL-1 blue cells (Agilent) by heat shock, and the cells were plated onto LB-agar plate containing 100 mg/L ampicillin at 37 °C for 24 h. A single colony on the agar plate was inoculated into 5 mL of LB media with 100 mg/L ampicillin, followed by incubation at 37 °C for 24 h. Plasmids in XL-1 blue cells were purified using Express plasmid Miniprep kit (Biomiga) and sequenced in Eton Bioscience. The sequenced plasmid was transformed into chemically competent *E. Coli* BL21(DE3) (New England Biolabs) cells containing a *ccm* (cytochrome *c* maturation) cassette with chloramphenicol resistance gene by heat shock. The cells were plated on LB-agar plate containing 100 mg/L ampicillin and 33 mg/L chloramphenicol at 37 °C for 24 h.²⁻³ 3–5 isolated colonies were incubated in 5 mL of LB media with 100 mg/L ampicillin and 33 mg/L chloramphenicol for 12–14 h. These cultures were inoculated into 2.8-L glass flasks containing 1.2 L of LB media (25 g/L LB, 100 mg/L ampicillin, and 33 mg/L chloramphenicol). Flasks were shaken at 100 rpm for 16–24 h at 37 °C. Pink-colored cells were harvested using centrifugation at 6,000×g for 10 min. The harvested cell pellets were stored at –80 °C.

Frozen cell pellets were dispersed in a 10 mM sodium acetate buffer solution (50–100 mL, pH 4.5). The pellets were fully thawed and vigorously stirred before centrifugation at 10,000 rpm for 10 min. Red-colored supernatant was stored in a plastic bottle at 4 °C. The precipitated pellets were frozen, thawed, and centrifuged twice to extract residual proteins from the periplasm. The pH of collected supernatants was adjusted to 10.0 using 1 M NaOH solution and immediately lowered to pH 4.5 using 50% (v/v) acetic acid solution to induce precipitation of impurities other than cyt *cb*₅₆₂ variants. After centrifugation (10,000 rpm for 10 min), the red-colored supernatant was filtered using 0.45 µm syringe disk-filters. The supernatant was introduced to a High-S cartridge column pre-equilibrated with 10 mM sodium acetate (pH 4.5). The loaded proteins were eluted using a step-gradient of 0–1000 mM NaCl with a Biologic DuoFlow workstation (BioRad). Fractions with A_{415nm}/A_{280nm} values > 5–6 were pooled, concentrated, and buffer-exchanged using 20 mM MOPS (pH 7.4) solution and a Vivaspinn-6 3kDa-centrifugal filter (Sartorius). To catalyze disulfide bond formation, CuCl₂ (0.5× protein monomer concentration) was incubated with the proteins at 37 °C for 8 h. The protein was treated with EDTA (10-fold molar excess of CuCl₂) at

4 °C to prepare metal-free protein. To remove EDTA, Cu^{II}-bound EDTA, and monomeric protein, the protein solution was exchanged six times (dilution factor 10⁶) using a Vivaspin-6 10 kDa-centrifugal filter (Sartorius) and a Chelex100-treated, 20 mM MOPS (pH 7.4) wash solution. The purity of the protein was evaluated with SDS-PAGE and A₄₁₅/A₂₈₀ ratio (> 6).⁴ The purified protein solution was metal-free and EDTA-free as determined by ESI-MS (Figure S2b) and ICP-MS analysis. The protein was concentrated to ~10 mM and stored at 4 °C.

X-ray Crystallography. Crystal growth conditions are presented in Table S2. B₂ crystals were obtained by sitting-drop vapor diffusion at room temperature. Prior to the crystallization, B₂ solutions were incubated with the metal ion of interest for 1 h. Under competitive conditions (Cu^{II} versus Zn^{II}), protein solutions were incubated with the first metal ion for 1 h. The second metal ion was added and incubated for additional 24 h. Sitting drop contained 1–2 μL each of the protein and precipitant solutions, and the reservoir contained 500 mL of the precipitant solution. Deep-red-colored crystals typically appeared within 4 weeks and were harvested after 2–8 weeks of maturation. For cryoprotection, except for 1Ni^{II}-B₂ crystal formed in MPD, perfluoro polyether (Hampton) was used for crystal freezing in liquid N₂. Diffraction data were collected at Advanced Light Source Beamline 5.0.2, and Stanford Synchrotron Radiation Lightsource Beamline 9-2 using multi-wavelength radiation at 100 K. All diffraction data were processed using iMosflm in CCP4 software suite.⁵ Molecular replacement was performed using Phaser-MR⁶ in the Phenix software package with monomeric cyt *cb*₅₆₂ (PDB: 2BC5) as a search model. Recursive model refinement using phenix.refine and metal/water placement using Coot⁷ were performed to generate final models. All graphics related to structures and electron densities were produced using PyMol.⁸ Data collection and refinement statistics are tabulated in Tables S3–4.

Electrospray Ionization Mass Spectrometry (ESI-MS). All ESI-MS experiments were performed using an Elite hybrid linear ion trap-orbitrap mass spectrometer equipped with a heated-electrospray ionization (HESI) source (Thermo). To remove non-volatile salts (e.g., MOPS buffer and NaCl) which suppress efficient ionization of proteins in ESI and non-specifically bound metal ions on the proteins, B₂ solutions were exchanged into a solution of 20 mM NH₄HCO₃. The concentrations of B₂ and metal ions in ESI-MS samples were adjusted to 5 μM and 15 μM in 20 mM NH₄HCO₃. The capillary voltage of ESI source was adjusted to 2.5–3.0 kV, the flow rate was set to 3–10 mL/min, the source temperature was set to 100 °C, and the sheath and auxiliary gas

flow rates were set to 12 (a.u.). Resolution ($m/\Delta m$) of the mass spectrometer (high mass mode & resolution 240,000) was estimated to $\sim 80,000$ around +11 charge state of B₂. To prevent metal contamination of ESI-MS samples during sample measurements, peek tubing and steel capillary were flushed out using 0.1% formic acid and equilibrated with 20 mM NH₄HCO₃ before the next sample measurement. 200 spectra were recorded for 3–4 min in LTQ tune software and averaged in QualBrowser software for further analysis. Theoretical and experimental m/z values of B₂ complexes observed in non-competitive and competitive conditions are listed in Tables S5–6, respectively.

Periplasmic Extraction of B₂. The extraction protocol was described in a previous study.⁹ Briefly, BL21(DE3) colonies with pET20b(+)/B₂ vector were transferred to 25 mL LB media in a 50 mL Falcon tube and incubated at 200 rpm for 16 h at 37 °C. Then, the cultures were further incubated with supplementary metal ions (50 μ M) at 200 rpm for 8 h at 37 °C. Cells were harvested using centrifugation at 8,000 $\times$ g (10 min and 4 °C). The cell pellets were washed with 4 mL of an ice-cold 1 $\times$ phosphate buffered saline (PBS) solution. The pellets were resuspended in 1.5 mL of a 20 mM NH₄HCO₃ solution with 20% (w/v) sucrose and transferred to 2 mL tubes. The suspension was incubated for 20 min at room temperature and centrifuged at 17,000 $\times$ g for 1 min. The pellet was then resuspended with 0.75 mL of ice-cold 20 mM NH₄HCO₃ solution to induce the lysis of the outer membrane. After incubating 20 min at room temperature, the cells were centrifuged at 17,000 $\times$ g for 2 min. The red-colored supernatant was diluted to 2 mL using a 20 mM NH₄HCO₃ solution and filtered using a Vivaspinn-6 100 kDa-centrifugal filter (Sartorius) to remove cell debris and large molecular weight components. The periplasmic extracts were filtered using an Amicon ultra-0.5 50 kDa-centrifugal filter (Millipore). The filtrate was concentrated with an Amicon ultra-0.5 30 kDa-centrifugal filter (Millipore) and buffer-exchanged using a 20 mM NH₄HCO₃ solution 4–6 times to remove low molecular weight species and salts. The A₄₁₅/A₂₈₀ ratio of the extracts was in the range of 2–4 after the filtration steps. Expression of B₂ in the extracts were identified using SDS-PAGE. Acrylamide gels (stacking/resolving gel percentage: 6%/15%) were stained using Coomassie blue staining or heme staining protocol. ESI-MS spectra of the purified extracts characterized in SDS-PAGE were obtained under the same parameters as above.

Density Functional Theory (DFT) Calculations. To examine the stability of trinuclear Zn^{II} sites of B₂ relative to Cu^{II}, DFT calculations were performed using Gaussian09. Two metal exchange

reaction models between $3\text{Zn}^{\text{II}}\text{-B}_2$ and $2\text{Zn}^{\text{II}}+1\text{Cu}^{\text{II}}\text{-B}_2$ were prepared using primary coordination spheres and metal ions observed in the crystal structure of $3\text{Zn}^{\text{II}}\text{-B}_2$. In Model 1, one of the Zn^{II} ions coordinated to 3His and 1Glu was replaced with Cu^{II} . In Model 2, the central Zn^{II} ion coordinated to 2His and $2\text{H}_2\text{O}$ was replaced with Cu^{II} . In both Models 1 and 2, free metal ions were described as $[\text{Cu}^{\text{II}}(\text{H}_2\text{O})]_2$ or $[\text{Zn}^{\text{II}}(\text{H}_2\text{O})]_2$ complex. Peptide backbones of metal-binding ligands and the rest of the protein structure were removed to reduce computational costs. Then, carbon-carbon covalent bonds which connect side chains and peptide backbone of the protein were replaced with carbon-hydrogen covalent bonds to meet the octet rule. Then, single point energy calculations were performed using the B3LYP¹⁰⁻¹¹ functional with the 6-31G+ (d,p) basis set¹²⁻¹⁴ in Gaussian09. Energy differences in the metal exchange reactions were evaluated, and Hartree energy values were converted to units of kcal/mol.

Inductively Coupled Plasma Mass Spectrometry (ICP-MS). Metal content analysis using ICP-MS was performed at the California NanoSystems Institute at the University of California, Los Angeles using NexION 2000 (PerkinElmer). All ICP-MS samples were prepared following an acid digestion protocol using hydrochloric acid and hydrogen peroxide. The concentrations of B_2 and metal ions were adjusted to 5 μM and 15 μM in 500 μL of 20 mM MOPS and 150 mM NaCl (pH 7.4). To remove non-specifically bound Cu^{II} or Zn^{II} on the protein surface, the protein solutions were exchanged three times using an Amicon ultra-0.5 10 kDa-centrifugal filter (Millipore) and 20 mM MOPS and 150 mM NaCl solution (pH 7.4). The concentrated protein solutions (~ 50 μL) were mixed with 50 μL of 1% (w/w) HCl and 50 μL of 33% (w/w) H_2O_2 . The mixtures were incubated at 60 $^\circ\text{C}$ for 3 h; the red color of cyt *cb*₅₆₂ disappeared after the acid treatment. The digested protein solutions were diluted to 5 mL using a 1% (w/v) HCl solution, and the final concentrations of metal ions were approximately within 10–1000 ppb, which corresponds to the range of linear response for ICP-MS analysis. The exogenous metal content was internally standardized to the heme-Fe content of each sample (~ 10 μM).

Competitive Metal Titration assays. Equilibration dissociation constants of metal-bound B_2 complexes were calculated using Fura-2, Mag-Fura-2, and Newport Green DCF^{1, 15-16} dyes that show changes in UV absorbance upon complexation with metal ions. Chelex100 resin (Sigma Aldrich) was utilized to remove trace amounts of metal ions in the buffer solution containing 20 mM MOPS and 150 mM NaCl (pH 7.4). Concentrations of all metal stock solutions were

determined using a PAR assay.¹⁷ To a 1-mL MOPS solution, 5 mM of metal-free dye and 5 mM of metal-free B₂ were added. Then, aliquots of 2 mM NiCl₂, CuCl₂ or ZnCl₂ stock solutions were added to induce the competition between the dyes and the proteins. Metal-induced changes in the absorbance spectra of the dyes were monitored with an Agilent 8453 UV-vis spectrophotometer at 25 °C. Complete equilibration was achieved within 5 min following metal addition. Changes in the absorbance at 335 nm (Fura-2), 324 nm (Mag-Fura-2), and 511 nm (Newport Green DCF) were used to measure the concentration of metal-bound dyes. Absorbance changes from three independent measurements were averaged as a single titration curve with the standard deviations. Then, the titration curves were fit to a one or two site binding model for protein using Dynafit.¹⁸

Solution Small Angle X-ray Scattering (SAXS). Experimental scattering profiles of B₂ were obtained at Beamline 4-2 of the Stanford Synchrotron Radiation Laboratory (SSRL). Concentrations of B₂ were adjusted to 5 mg/mL (200 μM dimer) in a 20 mM MOPS buffer solution (pH 7.4) containing 150 mM NaCl solution. Metal ion concentrations were fixed at 600 μM. B₂ solutions were prepared in 8×PCR strip tubes (VWR). The BL4-2 autosampler¹⁹ injected 30 μL of the solution into a quartz capillary tube with a diameter of 1.5 mm. Scattering patterns of the samples were recorded ten times (1 pattern/sec) using Pilatus3 X 1M Pixel Array detector (Dectris). There was no significant change in the patterns for ten measurements at 20 °C. Sample-to-detector distance was 1.7 m ($q < 0.5 \text{ \AA}^{-1}$).

SAXS Data Analysis. Experimental SAXS profiles were analyzed using ATSAS (v. 3.0.1) software package²⁰. The scattering profiles were normalized in SAS data analysis. Guinier analysis was performed using the equation below:

$$\ln[I(q)] = \ln[I(0)] - \frac{1}{3} q^2 R_g^2$$

where q and $I(q)$ are the scattering vector and the scattering intensity. Theoretical scattering profiles of 3Zn^{II}-B₂ and 1Ni^{II}-B₂ structures were generated in CRY SOL²¹ and compared with experimental scattering profiles. Chi-square (χ^2) values between the experimental and theoretical profiles were calculated using the equation below:

$$\chi^2 = \frac{1}{m-1} \sum_{k=1}^m \left(\frac{I_{theo}(q) - I_{exp}(q)}{\sigma_{exp}(q)} \right)^2$$

where m is the total number of points in scattering profiles, and σ_{exp} is the standard deviation of experimental scattering profiles. σ_{exp} values were obtained using repeat exposures of the same sample. Both theoretical and experimental scattering profiles were also converted to normalized Kratky plots using the function below to emphasize q -range from 0.1 to 0.3 Å⁻¹. This q -range corresponds to the change of dimer orientation.

$$(qR_g)^2 I(q)/I(0) \text{ vs. } qR_g$$

Analytical Ultracentrifugation. The oligomeric states of B₂ in metal-free and metal-bound states were characterized using analytical ultracentrifugation (AUC). The concentration of B₂ dimer was 10 μM in a buffer solution containing 20 mM MOPS and 150 mM NaCl (pH 7.4). Cu^{II} and Zn^{II} addition were made at three equivalents of metal ions per protein dimer. Sedimentation velocities of the protein solutions were measured on a Beckman XL-A instrument equipped with an An-60 Ti rotor. Samples were monitored at 435 nm (corresponding to the Soret band of the heme cofactor). For each sample, 600–900 scans were recorded at 41,000 rpm and 25 °C for 8–12 h using ProteomLab software. The scans were processed using SEDFIT software²² using with the following fitting parameters: $d_{\text{buffer}} = 1.0049 \text{ g ml}^{-1}$; $\eta_{\text{buffer}} = 0.010214 \text{ poise}$; $V_{\text{bar}} = 0.73084$. The fitting parameters were calculated by SEDNTERP. All results correspond to a confidence level of 95%.

Calculation of Metal-free and Metal-bound B₂ Fractions. Fractions of metal-free and metal-bound B₂ species were calculated using K_d values of 1Cu^{II}-B₂ and 3Zn^{II}-B₂ as a function of metal concentration. All mathematical models of the B₂ fractions were derived from equations for equilibrium dissociation constants. Fractions of individual B₂ species are represented as

$$F_{\text{nM}^{\text{II}}} = \frac{[\text{nZn}^{\text{II}}\text{-B}_2]}{[\text{B}_2]_{\text{tot}}}, F_{\text{Cu}^{\text{II}}} = \frac{[\text{Cu}^{\text{II}}\text{-B}_2]}{[\text{B}_2]_{\text{tot}}} - (1)$$

where $[\text{B}_2]_{\text{tot}}$ indicates the total amount of metal-free and metal-bound B₂ species, and n stands for the number of Zn^{II} ions bound to B₂. Using the definition of K_d , $[\text{B}_2]_{\text{tot}}$ can be expressed as follows:

$$[\text{B}_2]_{\text{tot}} = [\text{B}_2] + [\text{Cu}^{\text{II}}\text{-B}_2] + [\text{Zn}^{\text{II}}\text{-B}_2] + [2\text{Zn}^{\text{II}}\text{-B}_2] + [3\text{Zn}^{\text{II}}\text{-B}_2] - (2)$$

$$= [\text{B}_2] + \frac{[\text{B}_2][\text{Cu}^{\text{II}}]}{K_{\text{d,Cu}^{\text{II}}}} + \frac{[\text{B}_2][\text{Zn}^{\text{II}}]}{K_{\text{d1,Zn}^{\text{II}}}} + \frac{[\text{B}_2][\text{M}^{\text{II}}]^2}{K_{\text{d1,Zn}^{\text{II}}}K_{\text{d2,Zn}^{\text{II}}}} + \frac{[\text{B}_2][\text{M}^{\text{II}}]^3}{K_{\text{d1,Zn}^{\text{II}}}K_{\text{d2,Zn}^{\text{II}}}K_{\text{d3,Zn}^{\text{II}}}} - (3)$$

$$[3\text{Zn}^{\text{II}}\text{-B}_2] = \frac{[\text{B}_2][\text{Zn}^{\text{II}}]^3}{K_{\text{d1,Zn}^{\text{II}}}K_{\text{d2,Zn}^{\text{II}}}K_{\text{d3,Zn}^{\text{II}}}} - (4)$$

Incorporating (3) and (4) into (1) yields:

$$\therefore F_{3\text{ZnM}^{\text{II}}} = \frac{[3\text{Zn}^{\text{II}}\text{-B}_2]}{[\text{B}_2]_{\text{tot}}} = \frac{\frac{[\text{M}^{\text{II}}]^3}{K_{\text{d1,Zn}^{\text{II}}}K_{\text{d2,Zn}^{\text{II}}}K_{\text{d3,Zn}^{\text{II}}}}}{1 + \frac{[\text{Cu}^{\text{II}}]}{K_{\text{d,Cu}^{\text{II}}}} + \frac{[\text{Zn}^{\text{II}}]}{K_{\text{d1,Zn}^{\text{II}}}} + \frac{[\text{M}^{\text{II}}]^2}{K_{\text{d1,Zn}^{\text{II}}}K_{\text{d2,Zn}^{\text{II}}}} + \frac{[\text{M}^{\text{II}}]^3}{K_{\text{d1,Zn}^{\text{II}}}K_{\text{d2,Zn}^{\text{II}}}K_{\text{d3,Zn}^{\text{II}}}}} - (5)$$

Fractions of $[\text{B}_2]$, $[\text{Zn}^{\text{II}}\text{-B}_2]$, $[2\text{Zn}^{\text{II}}\text{-B}_2]$, and $[\text{Cu}^{\text{II}}\text{-B}_2]$ are derived in the same manner. The concentrations of Zn^{II} and Cu^{II} were considered as buffered species to calculate the fractions of metal-free and metal-bound B_2 species.

Table S1. Metal dissociation constants of B₂ variants. Competitive metal titration assays using Fura-2 were performed in a Chelex100-treated solution (pH 7.4) containing 20 mM MOPS and 150 mM NaCl. The dissociation constants in 20 mM NH₄HCO₃ were determined using Mag-Fura-2 (Zn^{II}) and Newport Green DCF (Cu^{II}). Errors are derived from non-linear least square fitting using Dynafit.

		20 mM MOPS + 150 mM NaCl (pH 7.4)			20 mM NH ₄ HCO ₃ (pH 7.8)	
		Ni ^{II}	Cu ^{II}	Zn ^{II}	Cu ^{II}	Zn ^{II}
B₂	<i>K</i> _{d1}	1.5 (±0.2) × 10 ⁻⁸	2.8 (±0.8) × 10 ⁻¹²	8.1 (±2.0) × 10 ⁻⁹	3.9 (±2.2) × 10 ⁻⁹	1.1 (±0.2) × 10 ⁻⁸
	<i>K</i> _{d2}			2.4 (±0.7) × 10 ⁻⁸		1.2 (±0.7) × 10 ⁻⁷
	<i>K</i> _{d3}			7.1 (±0.3) × 10 ⁻⁸		2.2 (±2.1) × 10 ⁻⁵
H60A B₂	<i>K</i> _{d1}		7.4 (±2.5) × 10 ⁻¹²			
H63A B₂	<i>K</i> _{d1}		2.0 (±0.5) × 10 ⁻¹²			
H104A B₂	<i>K</i> _{d1}		9.9 (±1.8) × 10 ⁻¹³			
	<i>K</i> _{d2}		2.6 (±0.8) × 10 ⁻¹²			

Table S2. Crystallization conditions for reported crystal structures.

Structure ^a	Metal		Beamline	Precipitant
3Zn ^{II} -B ₂	ZnCl ₂	6 mM	ALS 8.0.3.	PEG3350 7%, (NH ₄) ₂ SO ₄ 200 mM, MES 100 mM, pH 6
Cu ^{II} //Zn ^{II} -B ₂	CuCl ₂ //ZnCl ₂	6 mM//6mM	SSRL 9-2	PEG3350 24%, CHES 100 mM, pH 9
Zn ^{II} //Cu ^{II} -B ₂	ZnCl ₂ //CuCl ₂	6 mM//6mM	SSRL 9-2	PEG3350 25%, CHES 100 mM, pH 9
1Ni ^{II} -B ₂	NiCl ₂	4 mM	ALS 8.0.3.	MPD 45%, NaCl 200 mM, TRIS 100 mM, pH 8
B ₂	CuCl ₂	4 mM	ALS 8.0.3.	PEG3350 25%, (NH ₄) ₂ SO ₄ 200 mM, Tris 100 mM, pH 8.0

^aDimer concentration was 2 mM.

Table S3. X-ray data collection and refinement statistics for apo- and metal-bound B₂ structures.

	1Ni^{II}-B₂	3Zn^{II}-B₂	Apo-B₂
PDB ID	8DRD	8DRF	8DRJ
Data collection			
Resolution range (Å)	48.7 – 1.89 (1.958 – 1.89)	38.92 – 1.7 (1.761 – 1.7)	44.86 – 2.4 (2.486 – 2.4)
Space group	<i>P</i> 2 ₁ 2 ₁ 2 ₁	<i>P</i> 2 ₂ 2 ₁	<i>C</i> 2 ₁
Unit cell (Å, °)	42.09 55.43 101.97 90 90 90	39.50 52.12 58.51 90 90 90	89.32 48.02 45.12 90 96.18 90
Total reflections	39542 (3834)	27680 (2727)	14058 (1383)
Unique reflections	19771 (1917)	13841 (1364)	7510 (730)
Multiplicity	2.0 (2.0)	2.0 (2.0)	1.9 (1.9)
Completeness (%)	99.99 (100.0)	99.11 (99.85)	99.40 (99.32)
Mean I/sigma(I)	17.69 (5.78)	22.06 (8.37)	11.32 (6.72)
Wilson B-factor	18.62	16.42	23.97
R-merge	0.02575 (0.1109)	0.02514 (0.0754)	0.03868 (0.08279)
R-meas	0.03641 (0.1569)	0.03555 (0.1066)	0.05471 (0.1171)
R-pim	0.02575 (0.1109)	0.02514 (0.0754)	0.03868 (0.08279)
CC1/2	0.999 (0.956)	0.999 (0.989)	0.996 (0.971)
Refinement			
Reflections used in refinement	19771 (1917)	13841 (1362)	7509 (730)
Reflections used for R-free	1977 (191)	1377 (137)	751 (73)
R-work	0.1893 (0.2091)	0.2123 (0.2573)	0.1914 (0.2250)
R-free	0.2155 (0.2713)	0.2421 (0.3103)	0.2831 (0.3302)
Number of non-hydrogen atoms	1822	954	1729
macromolecules	1647	825	1629
ligands	96	46	96
solvent	79	83	4
Protein residues	214	107	214
RMS(bonds)	0.007	0.007	0.009
RMS(angles)	0.84	0.87	1.04
Ramachandran favored (%)	99.04	100.00	95.67
Ramachandran allowed (%)	0.48	0.00	1.92
Ramachandran outliers (%)	0.48	0.00	2.4
Rotamer	0.00	0.00	0.00

outliers (%)			
Clashscore	2.63	5.88	8.59
Average B-factor	20.66	20.48	25.66
macromolecules	21.05	19.81	25.48
ligands	14.66	15.99	28.96
solvent	19.88	29.60	19.58

Table S4. X-ray data collection and refinement statistics for B₂ structures formed under competitive Cu^{II}//Zn^{II} and Zn^{II}//Cu^{II} conditions.

	Cu ^{II} //Zn ^{II} -B ₂	Zn ^{II} //Cu ^{II} -B ₂
PDB ID	8DRL	8DRM
Data collection		
Resolution range (Å)	39.64 - 1.65 (1.709 - 1.65)	39.26 – 1.55 (1.605 – 1.55)
Space group	<i>P</i> 22 ₁ 2 ₁	<i>P</i> 22 ₁ 2 ₁
Unit cell (Å, °)	39.64 52.41 58.63 90 90 90	39.26 51.32 58.82 90 90 90
Total reflections	29844 (2910)	35200 (3394)
Unique reflections	15118 (1485)	17722 (1725)
Multiplicity	2.0 (2.0)	2.0 (2.0)
Completeness (%)	98.97 (99.33)	99.23 (98.07)
Mean I/sigma(I)	15.46 (5.03)	20.18 (4.07)
Wilson B-factor	18.17	20.73
R-merge	0.04037 (0.1227)	0.03811 (0.173)
R-meas	0.03709 (0.1736)	0.0539 (0.2446)
R-pim	0.04037 (0.1227)	0.03811 (0.173)
CC1/2	0.998 (0.961)	0.998 (0.916)
Refinement		
Reflections used in refinement	15118 (1485)	17721 (1724)
Reflections used for R-free	1510 (148)	1772 (172)
R-work	0.1807 (0.1978)	0.1932 (0.2534)
R-free	0.2015 (0.2280)	0.2285 (0.3037)
Number of non-hydrogen atoms	1003	999
macromolecules	827	814
ligands	46	46
solvent	130	139
Protein residues	107	107
RMS(bonds)	0.007	0.007
RMS(angles)	0.85	0.79
Ramachandran favored (%)	100	100.00
Ramachandran allowed (%)	0.00	0.00
Ramachandran outliers (%)	0.00	0.00
Rotamer	0.00	0.00

Outliers (%)		
Clashscore	1.16	3.01
Average B-factor	22.58	26.21
macromolecules	21.80	15.46
ligands	17.34	20.81
solvent	29.35	32.36

Table S5. Theoretical and observed m/z values of B₂ complexes under non-competitive metal binding conditions.

	Theoretical m/z	Observed m/z
B ₂	2262.55	2262.56
1Cu ^{II} -B ₂	2268.15	2280.09
1Zn ^{II} -B ₂	2268.31	2268.29
2Zn ^{II} -B ₂	2274.07	2274.10
3Zn ^{II} -B ₂	2279.84	2279.82
1Ni ^{II} -B ₂	2267.72	2267.74

Table S6. Experimental m/z values of B₂ complexes observed under competitive metal binding conditions between Zn^{II} and Cu^{II} or Ni^{II}.

	m/z (Zn ^{II} //Cu ^{II})	m/z (Cu ^{II} //Zn ^{II})	m/z (Ni ^{II} //Zn ^{II})	m/z (Zn ^{II} //Ni ^{II})
1M ^{II} -B ₂	2268.28	2268.28	2268.28	2268.28
2M ^{II} -B ₂	2274.10	2274.10	2274.10	2274.10
3M ^{II} -B ₂	2279.72	2279.72	2279.82	2279.82

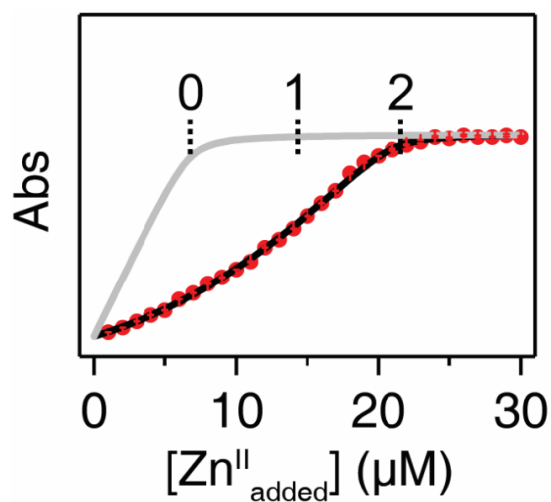


Figure S1. Competitive Zn^{II} titration between $(\text{AB})_2$ and Fura-2 in a 20 mM NH_4HCO_3 solution (pH 7.8). Changes in Fura-2 absorbance at 335 nm (red data points, black fit) is plotted along with the theoretical metal-binding isotherm (grey) in the absence of $(\text{AB})_2$. Simulated equivalence points (0–2 equivalents) for Zn^{II} binding to $(\text{AB})_2$ are shown on the theoretical metal-binding isotherm. Experimental data points and error bars are presented as mean and standard deviation of three independent measurements.

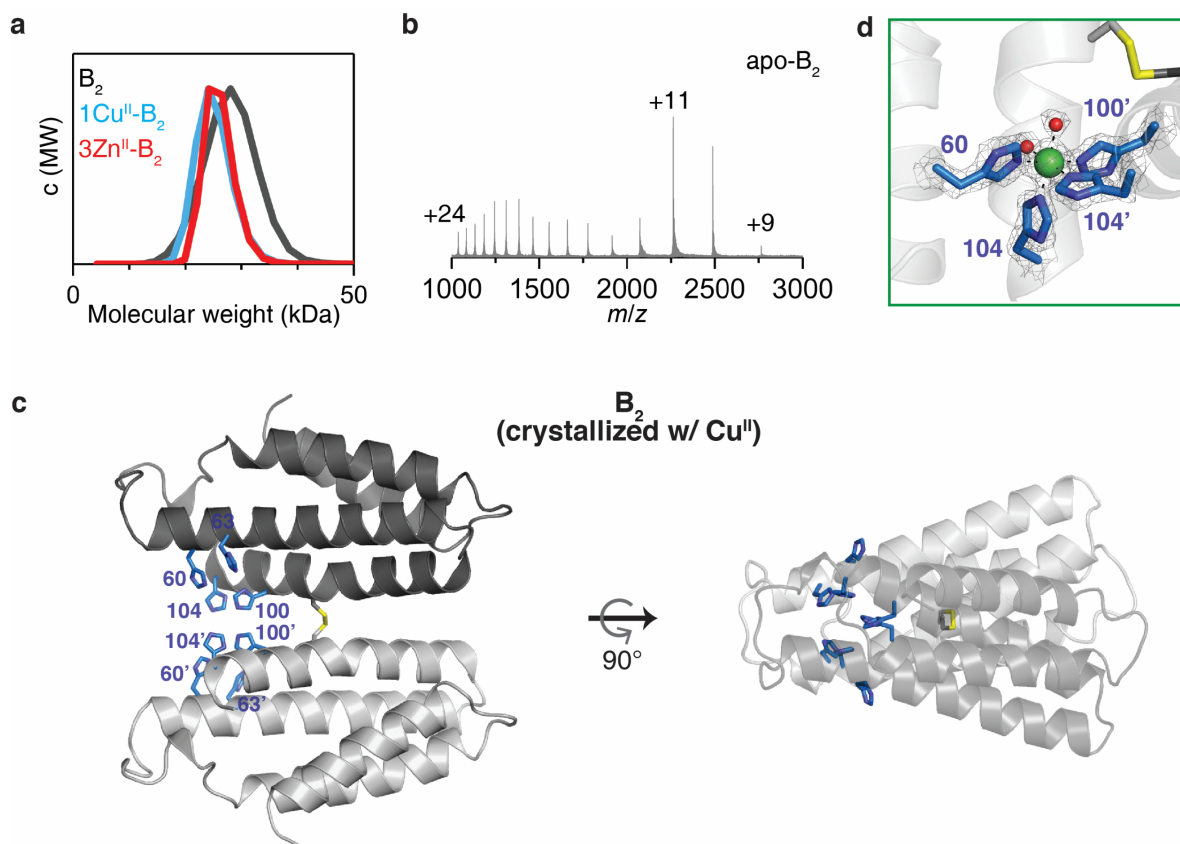


Figure S2. (a) Sedimentation velocity/AUC analysis of the oligomerization states of apo-B₂ (black), 1Cu^{II}-B₂ (blue), and 3Zn^{II}-B₂ (red) at 10 μ M B₂ dimer concentration. C(MW)_{max} of apo-B₂, 1Cu^{II}-B₂, and 3Zn^{II}-B₂ were observed at 28064 Da ($S_{\text{max}} = 2.6$ S), 24100.6 Da (2.4 S), and 24260.5 Da (2.4 S), respectively. The theoretical molecular weight of the B₂ dimer is 24878 Da. (b) Charge state distribution of apo-B₂ in ESI-MS spectrum. The +11 charge state was selected as the representative state for metal-binding analysis owing to its relatively low charge and high abundance. (c) Crystal structure of apo-B₂ crystallized in the presence of three molar equivalents of Cu^{II} (PDB ID:8DRJ). Blue sticks represent B site including the native H63 residue. (d) Close-up view of the primary coordination sphere of tetra-His Ni^{II} binding site. Green and red spheres represent Ni^{II} ion and water molecules, respectively. Contour levels of $2mF_0-DF_c$ (gray mesh) electron density are 5.0 σ for Ni^{II} and 2.0 σ for C, N, and O atoms.

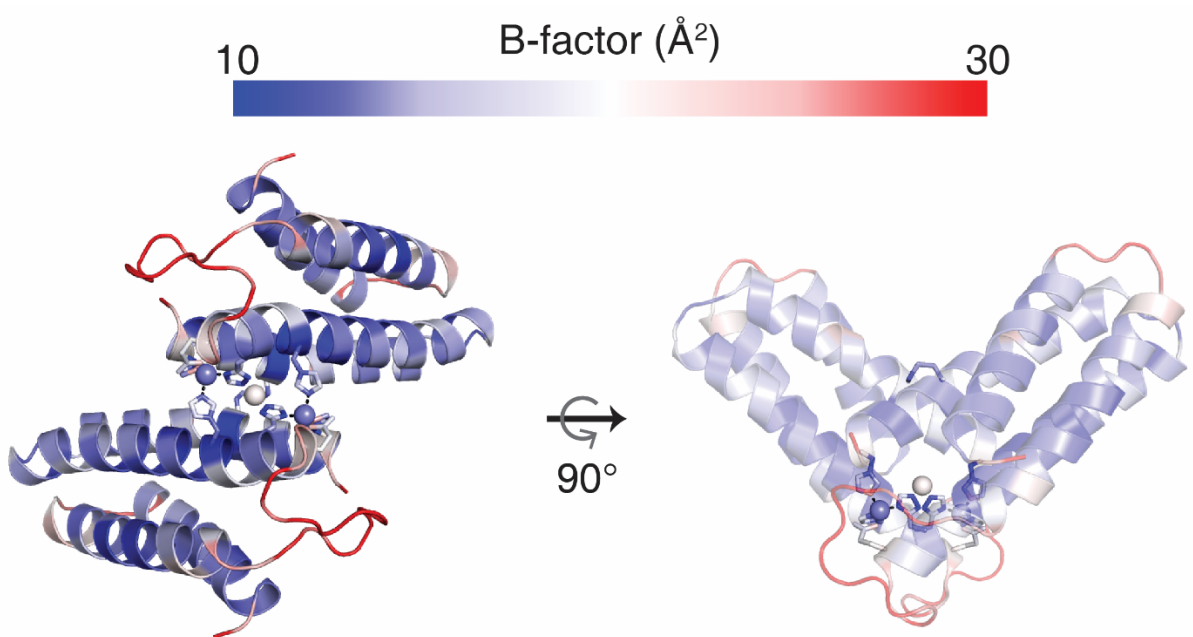


Figure S3. Cartoon representations of the $3\text{Zn}^{\text{II}}\text{-B}_2$ crystal structure colored according to B-factors (scale shown above). The average B-factors of H60, H63, H104, and E57 ($\sim 17.0 \text{ \AA}^2$) are smaller than that of the crystal structure ($\sim 20.5 \text{ \AA}^2$). The B-factor of the rigidified dimer interface (residues 57–68 and residues 97–104, $\sim 17.5 \text{ \AA}^2$) was also smaller than the average.

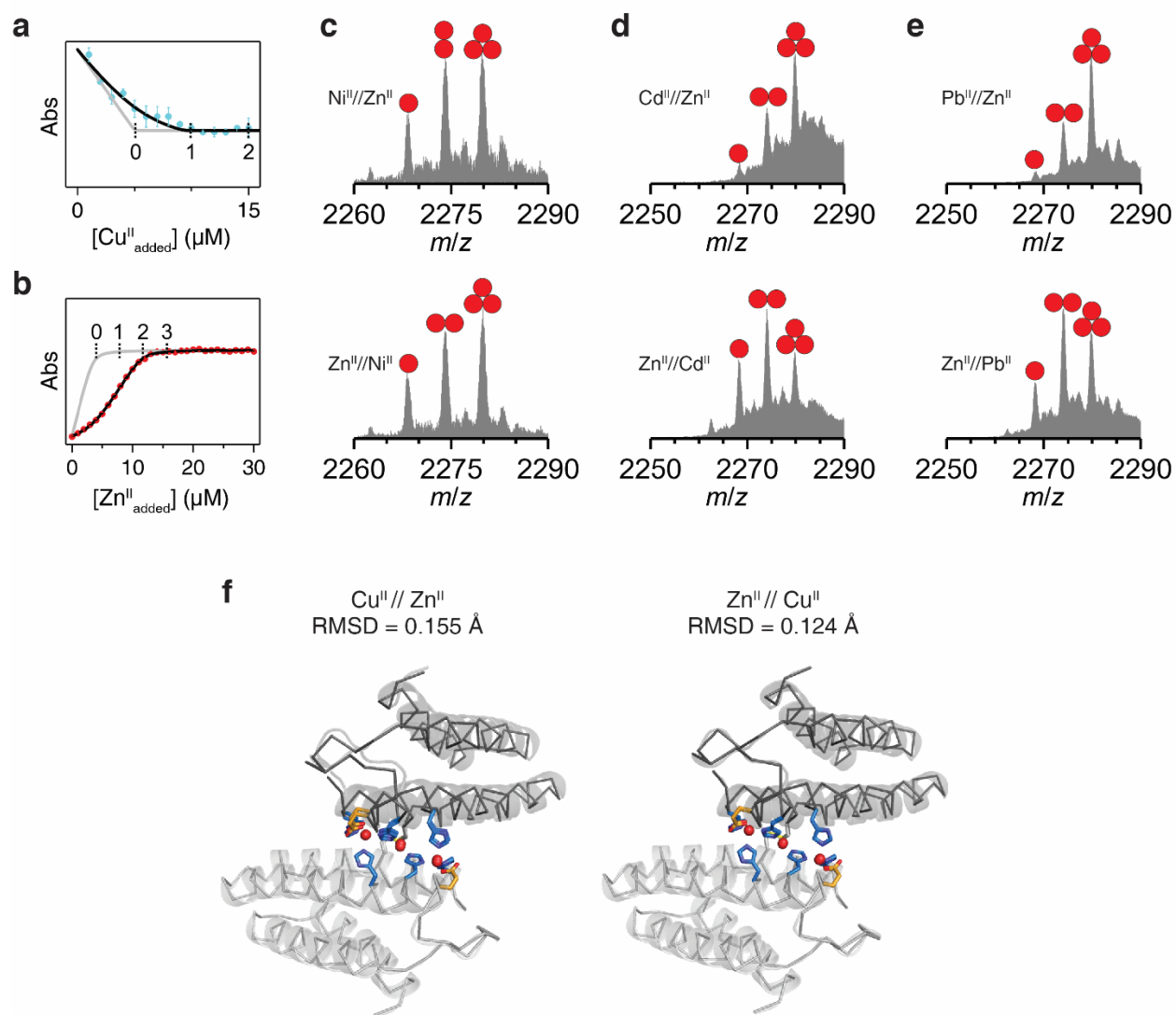


Figure S4. Competitive (a) Cu^{II} titration between B₂ and Newport Green DCF or (b) Zn^{II} titration between B₂ and Mag-Fura-2 in 20 mM NH₄HCO₃ (pH 7.8). Changes in Fura-2 absorbance at 335 nm (cyan or red data points and black fits) are plotted along with theoretical metal-binding isotherms (grey) in the absence of B₂. Simulated equivalence points (0–3 equivalents) for M^{II} binding to B₂ are indicated on the theoretical metal-binding isotherms. Experimental data points and error bars are presented as mean and standard deviation of three independent measurements. (c–e) ESI-MS spectra of B₂ obtained in (c) 20 mM NH₄HCO₃ (pH 7.8) and (d–e) 20 mM NH₄Ac (pH 6.8). The red circles in ESI-MS spectra represent the number of Zn^{II} ion bound to B₂. (f) Crystal structures (ribbon models) of B₂ formed in Cu^{II}//Zn^{II} (PDB:8DRL) and Zn^{II}//Cu^{II} (PDB:8DRM) conditions. Root mean square deviation (RMSD) values were determined in comparison with the crystal structures of 3Zn^{II}-B₂ (gray cartoon).

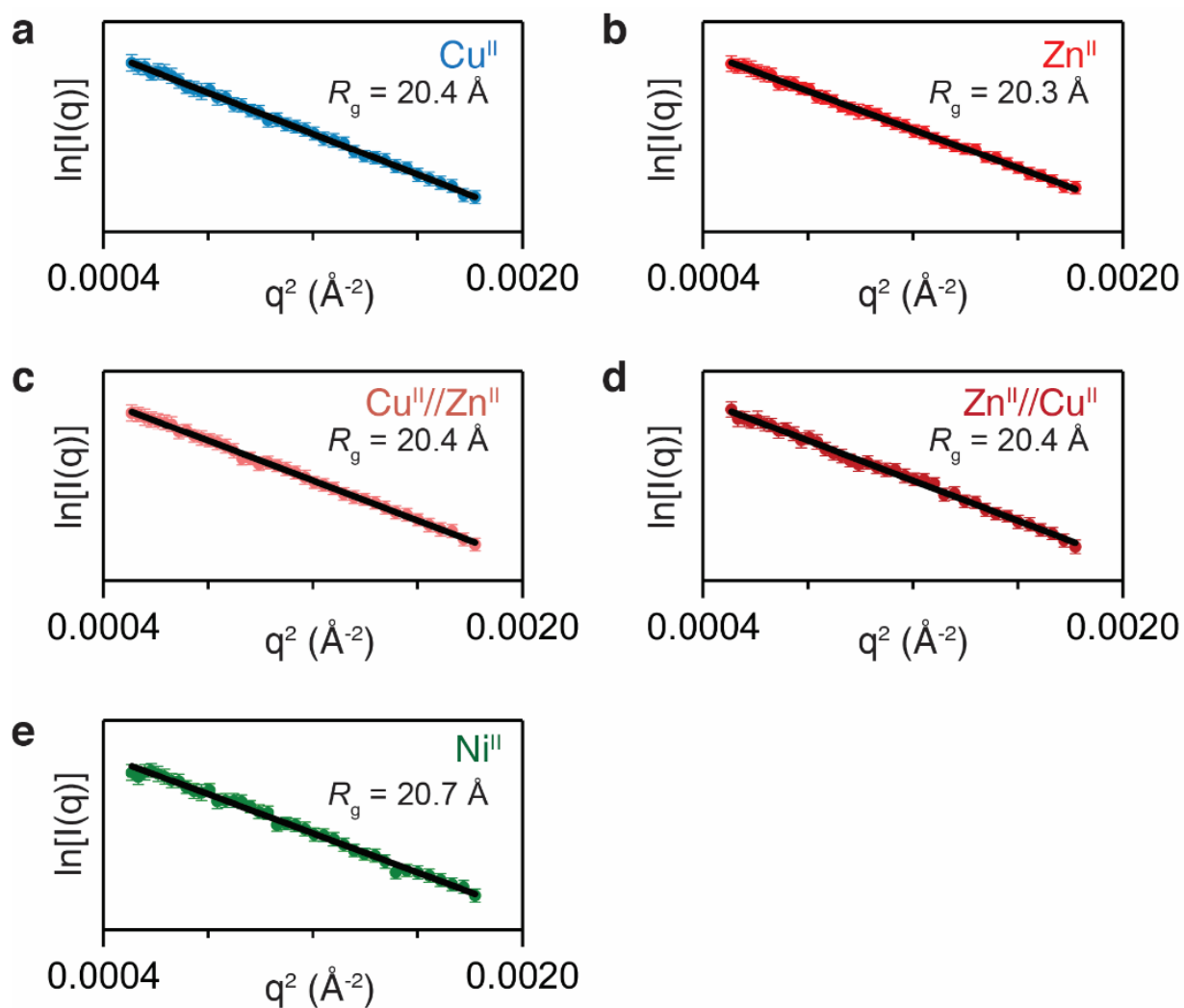


Figure S5. Guinier analysis of B₂ in (a) Cu^{II} only, (b) Zn^{II} only, (c) Cu^{II}//Zn^{II}, (d) Zn^{II}//Cu^{II}, and (e) Ni^{II} only conditions (200 μM protein dimer + 600 μM metal ions).

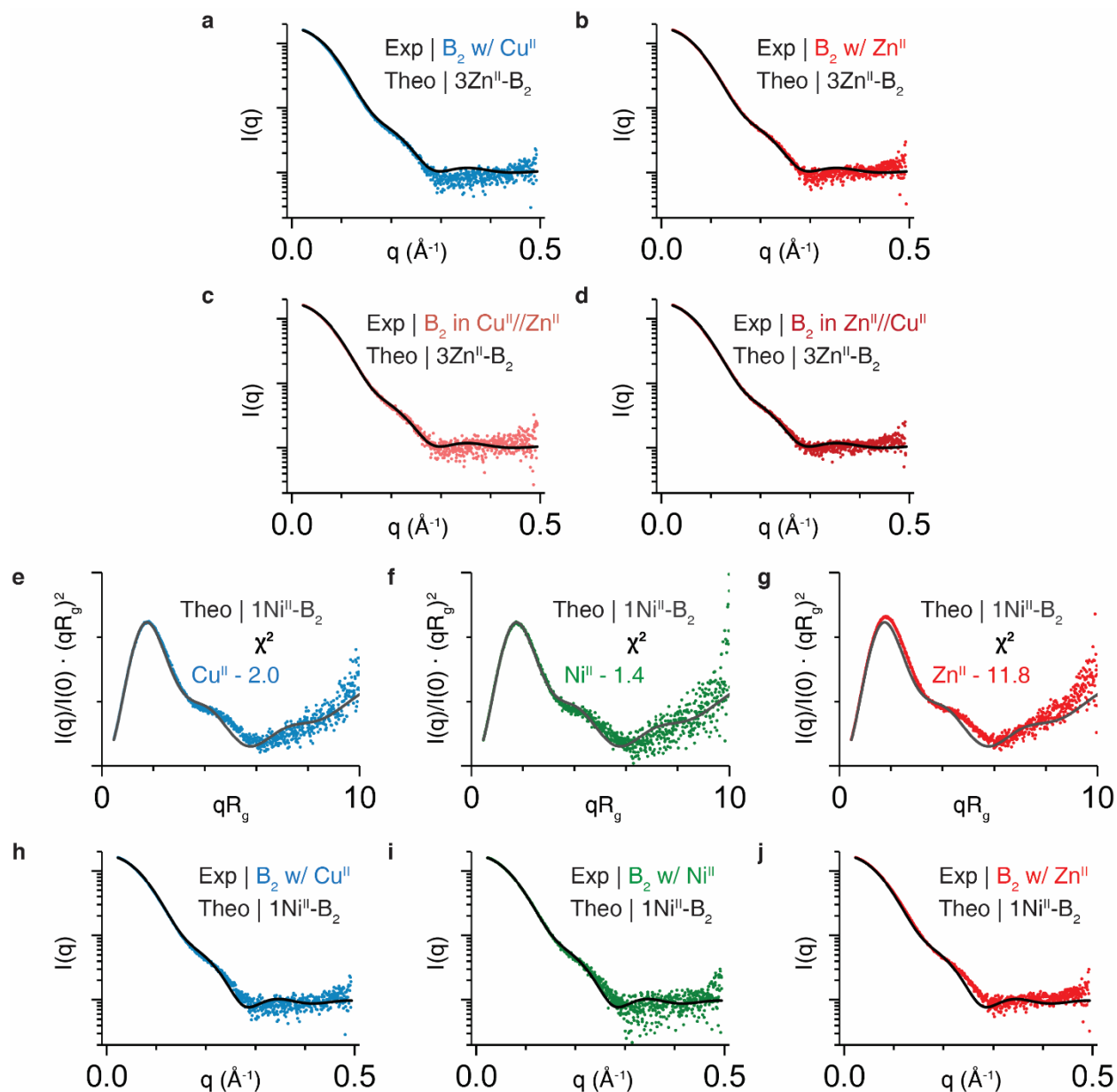


Figure S6. (a–d) Experimental SAXS profiles of B₂ in (a) Cu^{II} only, (b) Zn^{II} only, (c) Cu^{II}//Zn^{II}, and (d) Zn^{II}//Cu^{II} conditions at 200 μM dimer concentration ($[M^{II}] = 600 \mu\text{M}$, and $[M_1^{II}] // [M_2^{II}] = 600 // 600 \mu\text{M}$). (e–g) Normalized Kratky plots of B₂ in (e) Cu^{II} only, (f) Ni^{II} only, and (g) Zn^{II} only conditions compared with the theoretical profile of 1Ni^{II}-B₂ (black). (h–j) Experimental SAXS profiles of B₂ in (h) Cu^{II} only, (i) Ni^{II} only, and (j) Zn^{II} only conditions at 200 μM dimer concentration ($[M^{II}] = 600 \mu\text{M}$).

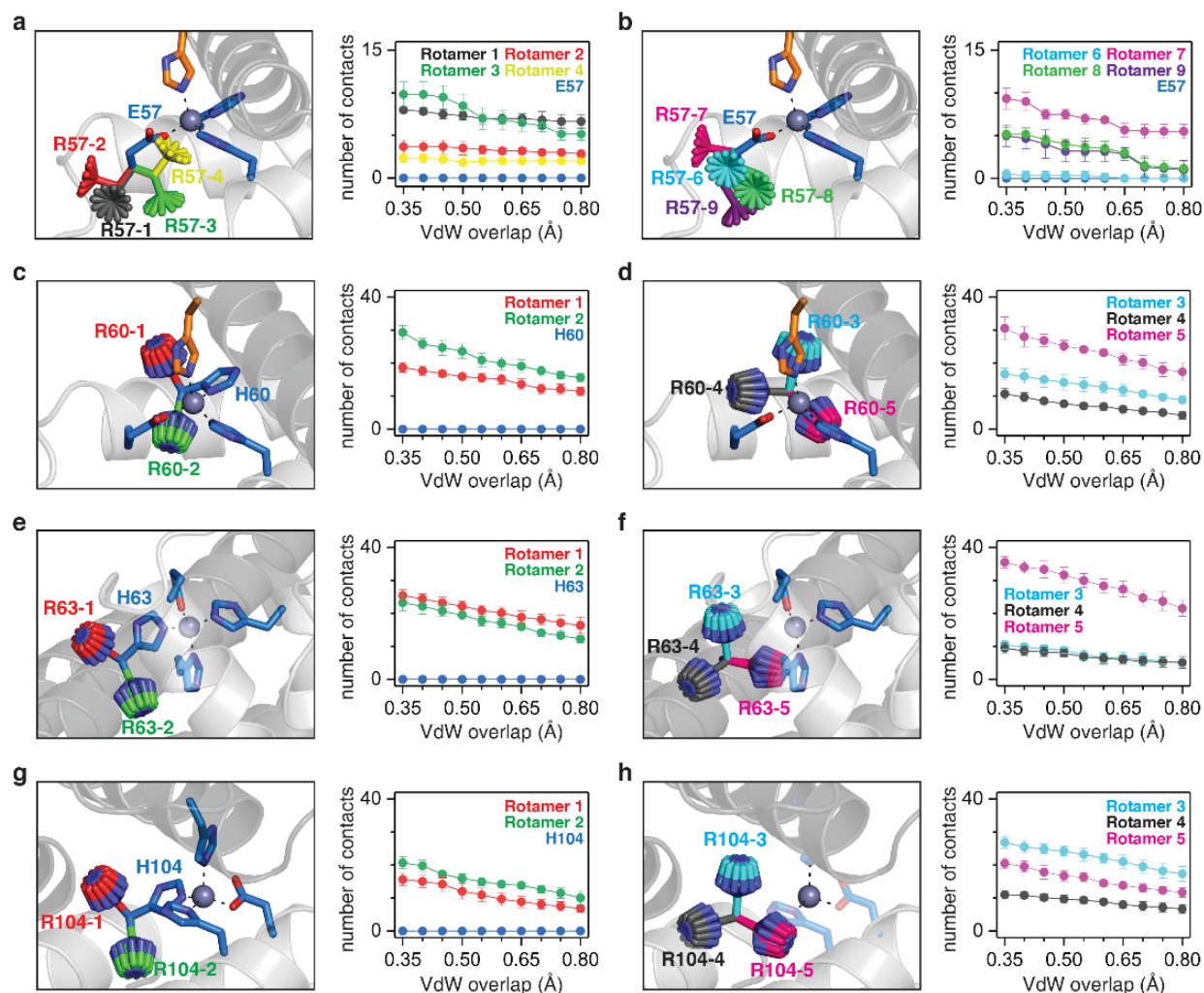


Figure S7. (a–h) Possible rotamer orientations (left) and average number of atomic contacts (right) of (a–b) E57, (c–d) H60, (e–f) H63, and (g–h) H104 residues in the crystal structure of 3Zn^{II}-B₂. Rotamers were generated with the combination of c_1 (torsion angle of $C_\alpha-C_\beta = 60-300^\circ$, 60° interval) and c_2 (torsion angle of $C_\beta-C_\gamma = 0-330^\circ$, 30° interval). The average number of contacts were presented as a function of van der Waals (vdW) overlap. Error bars reflect the standard deviations of the atomic contacts in rotamers with fixed c_1 and variable c_2 . Compared to the original conformations of E57, H60, H63, and H104, all possible rotamers show significant clashes with proximal residues.

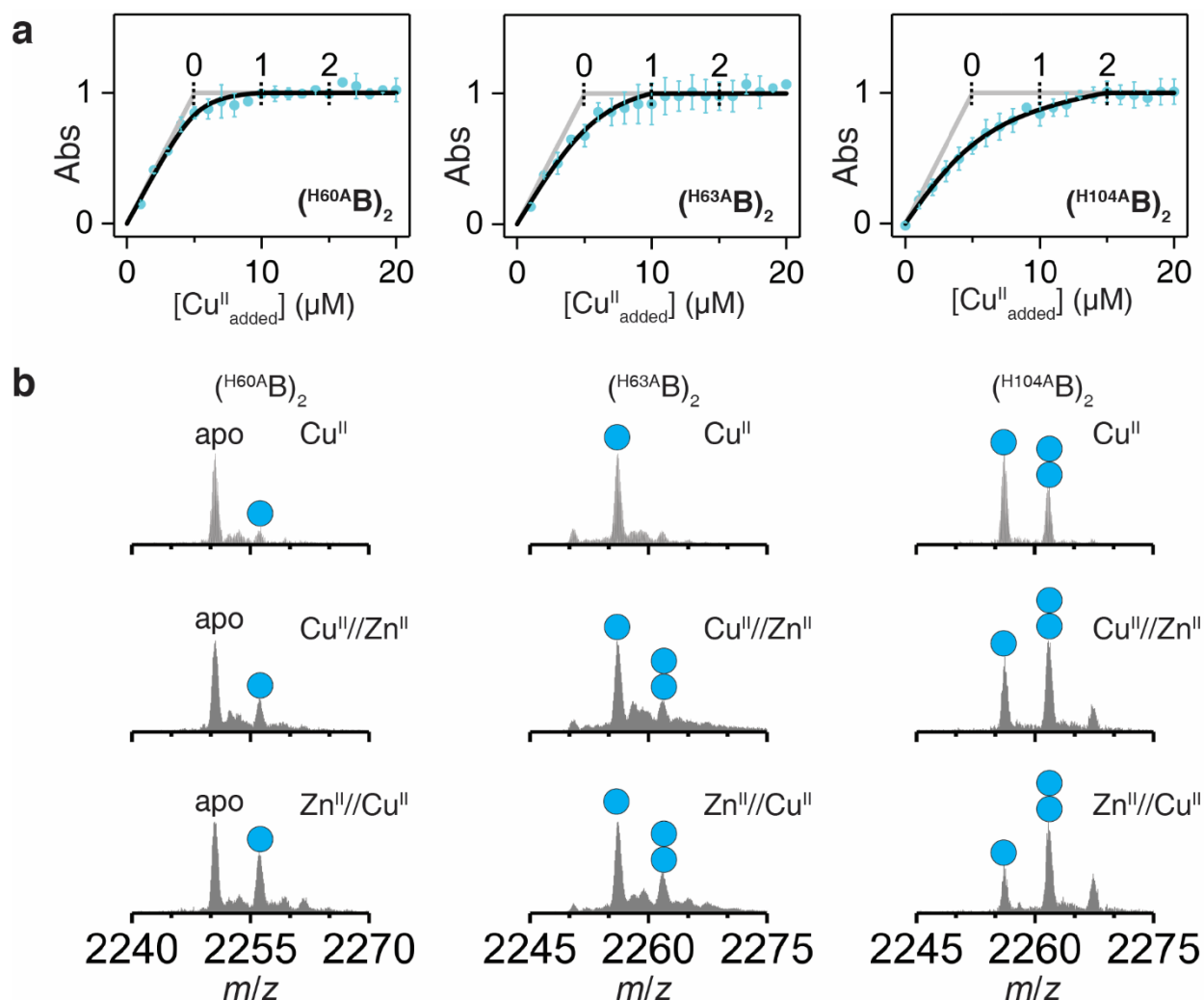


Figure S8. (a) Competitive Cu^{II} titration assays between B_2 mutants (H60A, H63A, and H104A), and Fura-2. Changes in normalized Fura-2 absorbance at 335 nm (cyan data points, black fits) are plotted along with theoretical metal-binding isotherms (grey) in the absence of B_2 . Simulated equivalence points (0–2) for Cu^{II} binding to B_2 are indicated with the theoretical metal-binding isotherms. Experimental data points and error bars are presented as mean and standard deviation of three independent measurements. (b) ESI-MS spectra of B_2 mutants (H60A, H63A, and H104A) under Cu^{II} only, $\text{Cu}^{\text{II}}/\text{Zn}^{\text{II}}$, and $\text{Zn}^{\text{II}}/\text{Cu}^{\text{II}}$ conditions. The cyan circles in ESI-MS spectra represent the number of Cu^{II} ion bound to B_2 mutants.

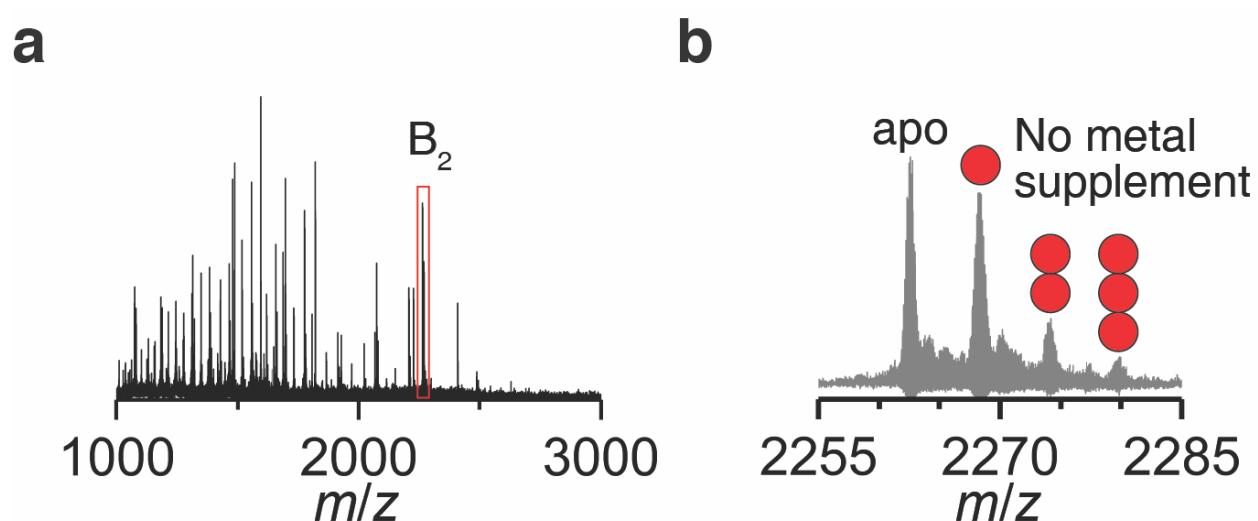


Figure S9. (a) ESI-MS spectrum of a periplasmic extract of *E. coli* grown without supplemental metal ions. Peak location of B_2 and its metal-bound complexes was spotted in the red box. (b) Expanded spectrum of the red box in panel (a). The red circles represent the number of Zn^{II} ion bound to B_2 . Even though no metal ion was supplemented to the media, naturally existing Zn^{II} ions ($\sim 17 \mu M$) in LB media induced metalation of B_2 in periplasm.

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